

Gaussian Image Steganography via Parameter-Domain Keyed Embeddings

Hide information in Gaussian parameters — not pixels

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2D Gaussian representation

Parameter-domain payload

Low-distortion embedding

Key-dependent decoding



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Hide information in Gaussian parameters — not pixels

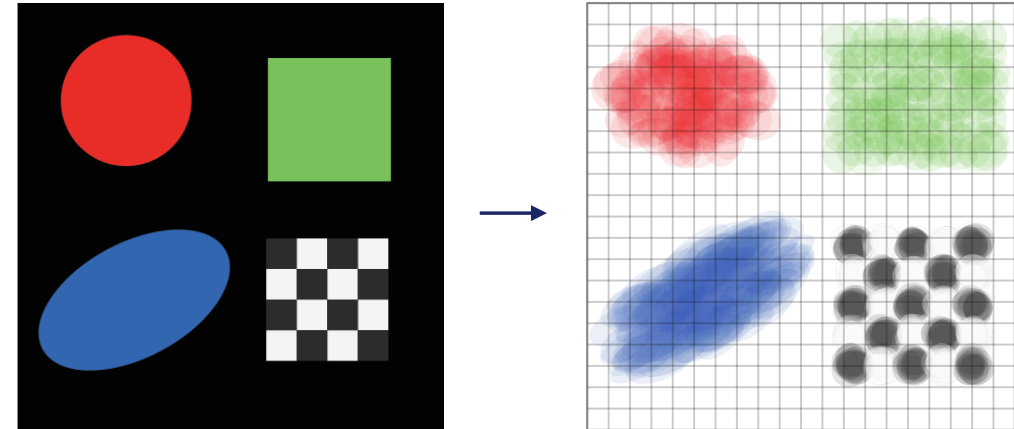
An image can be represented by a set of image-space 2D Gaussian primitives.

Each Gaussian contains parameters such as:

position · shape · orientation · color · opacity

We ask:

Can the Gaussian parameter set itself carry a hidden message while preserving a nearly unchanged rendering?



Clean Gaussian parameters

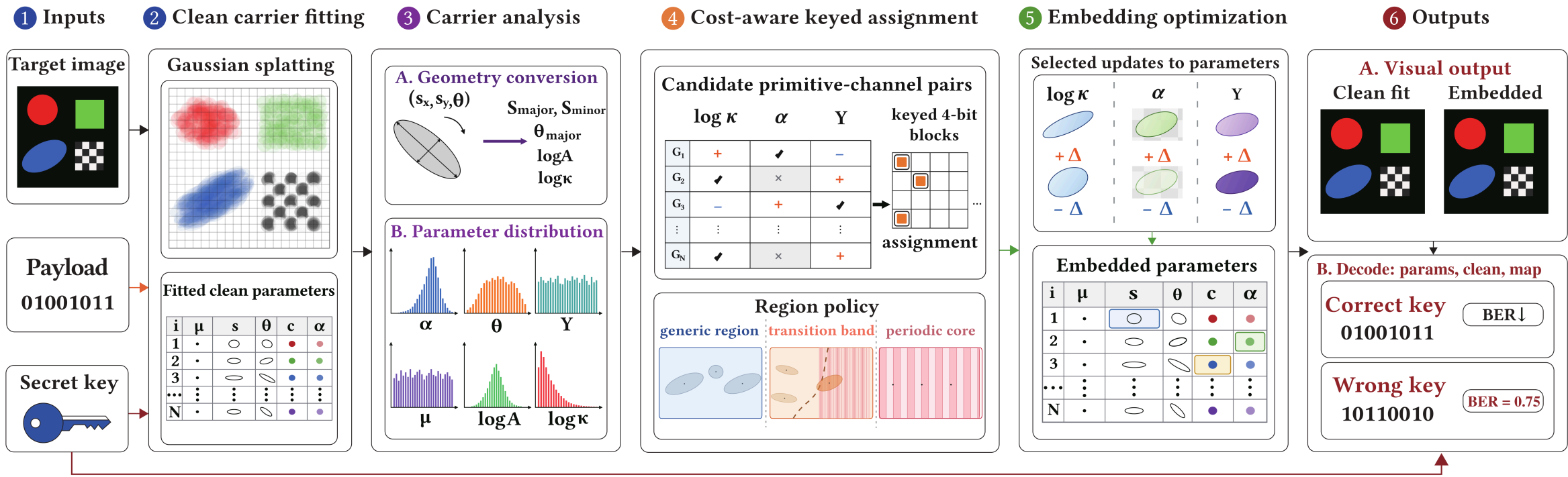
$$P = \{G_i\}_{i=1}^N$$

$$P \longrightarrow \tilde{P}$$

Small selected parameter changes
→ embedded Gaussian parameters

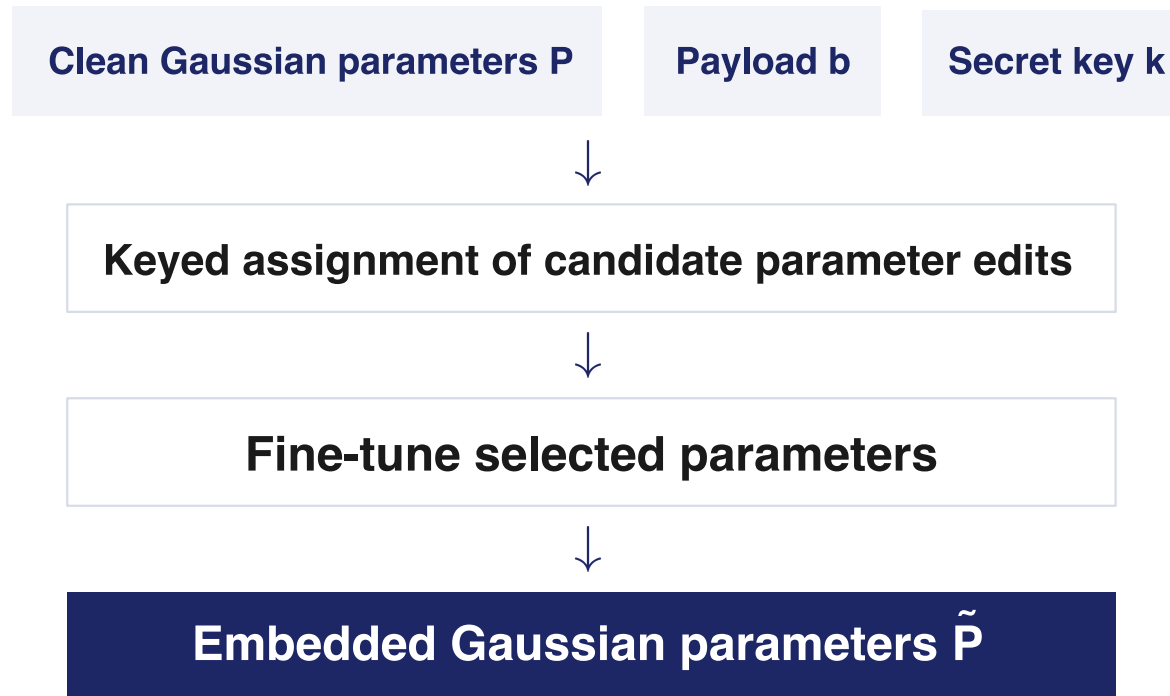
System overview: from clean fit to keyed recovery

Fit a clean carrier, select low-cost keyed edits, and recover the payload from parameter changes.



The key selects edits; decoding reads changes against the clean fit

Encoding



Decoding

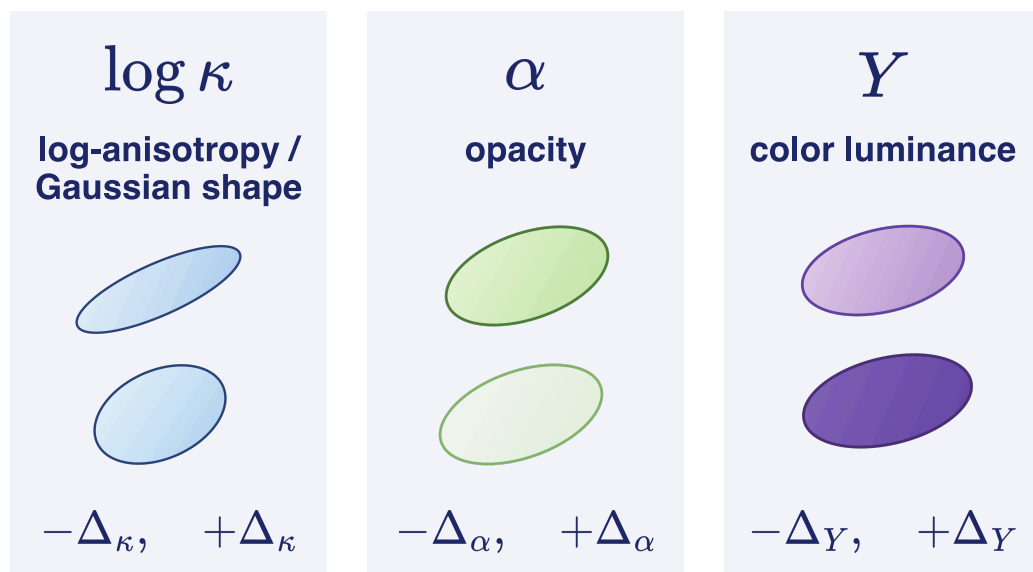
The decoder uses:

- embedded parameters $\tilde{P}$
- corresponding clean parameters P
- encoder-generated assignment map M

$$(\tilde{P}, P, M) \rightarrow \hat{b}$$

Three carrier channels, cost-aware keyed assignment

We construct candidate signed edits in three parameter channels:



The secret key determines the candidate signed actions and their mapping to 4-bit message blocks.

For each payload block, we select a valid assignment using:

- edit cost
- key-induced payload constraints
- region and coverage constraints

The selected parameters are then fine-tuned using objectives for:

- payload recovery
- rendered-image fidelity
- parameter regularization

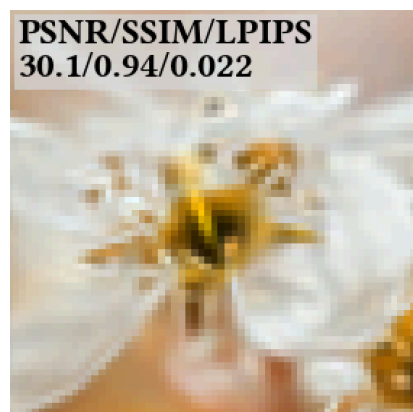
candidate edits $\rightarrow$ cost-aware keyed assignment $\rightarrow$ selected parameters

**Assignment determines where and how to edit.
Fine-tuning makes the selected edits recoverable while preserving the rendering.**

Small parameter edits keep the rendering close to the clean fit

For representative examples:

Source image → Clean Gaussian fit → Embedded Gaussian fit



The clean and embedded reconstructions remain visually similar.

The reported reconstruction metrics are measured against the source image.

The additional distortion introduced specifically by embedding is evaluated separately using:

$$\Delta\text{PSNR}$$

between the clean and embedded fits.

Fixed 8-bit payload: controlled fits and held-out natural images

Controlled synthetic evaluation

3 synthetic targets · 30 independent Gaussian fits
256 Gaussians · fixed 8-bit payload

Method	Key-dependent	Cost-aware	Exact recovery	Mean Δ PSNR ↓
Matched random	Yes	No	25 / 30	0.066 dB
Greedy no-key	No	Yes	30 / 30	0.075 dB
Ours	Yes	Yes	30 / 30	0.091 dB

p = 0.031 Against matched random selection
paired one-sided permutation test.

Cost-aware keyed assignment removes the five recovery failures observed in these 30 fits.

Held-out natural-image evaluation

256×256 images · 4,096 Gaussians
fixed 8-bit payload

112 / 112 exact payload recovery

0.009 dB
mean Δ PSNR

0.520
mean wrong-key BER

Matched random **110 / 112**

Greedy no-key **112 / 112**

4,096 Gaussians selected on a separate 12-image set;
the remaining 112 images were held out for evaluation.

Boundary tests clarify what the current method supports

The payload depends on the fitted parameter representation

After 200 additional reconstruction-only optimization steps:

$$\text{BER}_{\text{correct key}} = 0.264$$

After:

embedded parameters → render image
→ fit a new Gaussian representation

the correct-key BER becomes approximately:

$$\text{BER}_{\text{correct key}} \approx 0.75$$

The current payload is not preserved by rendering and refitting a new Gaussian representation.

Payload-length development experiment

Separate nine-fit development set:

Requested payload	Exact recovery
4 bit	8 / 9
8 bit	9 / 9
12 bit	5 / 9
16 bit	not demonstrated

All nine 16-bit requests were shortened to 8 bits by the fallback procedure.

These experiments identify 8 bits as the most reliable tested operating point.

Gaussian parameters can carry a small keyed payload



Parameter-domain

Information is embedded through selected changes to an image-space 2D Gaussian parameter representation.



Keyed + cost-aware

The secret key constrains valid assignments, while edit costs guide which parameter changes are used.



112 / 112

The fixed 8-bit payload recovered exactly on all 112 tested held-out natural images, with a mean PSNR drop of 0.009 dB.

Decoder requirements

embedded parameters + paired clean parameters + assignment map

Thank you

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BibTeX

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